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First documented occurrence and evidence of local establishment of the non-indigenous crab *Percnon gibbesi* at Theotokos, southeastern Pelion, Greece

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Abstract

The non-indigenous nimble spray crab *Percnon gibbesi* (H. Milne Edwards, 1853) has expanded rapidly across the Mediterranean Sea, including the Aegean and Ionian Seas, since its first regional record in 1999. We report the first documented occurrence of the species at Theotokos, southeastern Pelion, central Aegean Sea, Greece. An initial coastal biodiversity survey conducted under the NATIV-PELION monitoring programme recorded and photographed two individuals on 21 July 2026. A targeted follow-up survey on 22 July 2026 recorded 32 individuals at and around 39.19358894° N, 23.34314461° E, spanning size classes from small juveniles to large adults. Several specimens occupied crevices and sheltered interstices within the shallow rocky substrate, forming a dense local aggregation consistent with a probable settlement and refuge site. The co-occurrence of multiple size classes indicates that the record reflects more than the arrival of one or two transient individuals, and instead constitutes preliminary evidence of an established or actively establishing local population. Quantitative follow-up surveys are needed to characterise population density, spatial extent, reproductive status, and potential ecological interactions with native shallow-water grazers and benthic assemblages. This record extends the known local distribution of *P. gibbesi* into southeastern Pelion. It demonstrates the value of rapid follow-up surveys in revealing the true extent of a population following its initial detection.

Keywords: Aegean Sea, alien species, biological invasion, Brachyura, early detection, range expansion

Introduction

The nimble spray crab *Percnon gibbesi* is a primarily algivorous brachyuran native to tropical and subtropical waters of the Atlantic and eastern Pacific. First recorded in the Mediterranean Sea in 1999, it has since undergone rapid range expansion across much of the basin. The species characteristically inhabits shallow infralittoral rocky habitats, particularly substrates composed of boulders, cobbles and stones offering crevices and limited algal cover (Katsanevakis et al., 2010; Thessalou-Legaki et al., 2006; Deudero et al., 2005; Zenone et al., 2016).

The earliest confirmed Greek records of *P. gibbesi* date from 2004–2005, with initial observations at Antikythira, the Messiniakos Gulf, Crete and Rhodes, followed by further records from the

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Saronikos Gulf, Chios, Milos, Zakynthos, Syvota and other Aegean and Ionian localities (Katsanevakis et al., 2010; Thessalou-Legaki et al., 2006). The species was subsequently considered established at several of these sites, based on multiple individuals of varying sizes observed on shallow rocky substrates (Thessalou-Legaki et al., 2006).

Dispersal of *P. gibbesi* within the Mediterranean likely involves more than one mechanism. Vessel-mediated transport has been proposed as a plausible initial pathway of introduction, while subsequent spread between localities may be facilitated by planktonic larval dispersal and prevailing current patterns (Katsanevakis et al., 2010; Thessalou-Legaki et al., 2006). The species' invasion success is further supported by ecological and behavioural traits that favour rapid colonisation of shallow rocky habitats: experimental evidence indicates that late larval stages preferentially and rapidly select hard, stable substrates that offer interstitial shelter, such as cobbles and flat stones (Zenone et al., 2016).

Local-scale records remain valuable because the distribution of non-indigenous marine species is often incompletely documented outside intensively monitored harbours, islands and established research sites. Observations spanning multiple size classes are particularly informative, as they can provide preliminary evidence of recruitment and population persistence rather than isolated, transient occurrence.

This field note documents the first known occurrence of *P. gibbesi* at Theotokos, southeastern Pelion, Greece, describing an initial observation of two individuals on 21 July 2026 and a subsequent, markedly larger detection of 32 individuals spanning juvenile to adult size classes during a targeted survey on 22 July 2026.

Materials and methods

Study area

Observations were conducted at Theotokos, southeastern Pelion, central Greece, in the western Aegean Sea. The principal aggregation was located at and around the following coordinates:

39.19358894° N, 23.34314461° E

The surveyed coastal habitat comprised shallow rocky substrate with stones, boulders and interstitial refuges. Depth range, total surveyed area and dominant benthic cover were not quantified during this preliminary survey and should be recorded systematically in subsequent fieldwork.

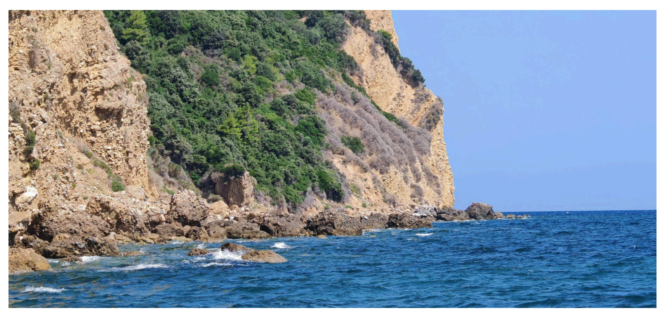

Fig.1: Location of the newly documented *Percnon gibbesi* population at Theotokos, southeastern Pelion, Greece. The principal aggregation was observed around 39.19358894° N, 23.34314461° E.

Initial observation

On 21 July 2026, a marine biodiversity survey was conducted at Theotokos under the NATIV-PELION programme of Merman Conservation Expeditions Ltd. Two individuals (Fig. 2) exhibiting the characteristic external morphology of *P. gibbesi* were observed and photographed in situ.

Fig.2: One of the first two *Percnon gibbesi* individuals photographically documented at Theotokos on 21 July 2026.

Identification was based on the conspicuously flattened body, elongate walking legs, pointed anterolateral carapace margins, and the colour pattern characteristic of *P. gibbesi*. Photographs were subsequently reviewed to verify the field identification.

Follow-up survey

A targeted follow-up search was conducted on 22 July 2026 within the area of the initial observation.

The observer systematically examined exposed rock surfaces, spaces beneath and between stones, crevices and other shallow-water refuges.

All separately visible individuals were counted, with the site surveyed progressively and the position of occupied shelters noted to minimise the risk of repeated counts. Individuals were not captured, handled or physically measured.

Specimens were assigned visually to broad relative size classes, as follows:

• Small juveniles

• Intermediate-sized individuals

• Large individuals or probable adults

These categories represent qualitative field estimates and should not be interpreted as formal carapace-length measurements.

Terminology

The concentrated occurrence of individuals within rocky shelters is described here as an aggregation, refuge site or probable settlement site. The term "nest" is deliberately avoided, as nest construction, egg deposition, parental care and occupation by ovigerous females were not confirmed during the survey.

Results

First observation

Two individuals of *Percnon gibbesi* were recorded at Theotokos on 21 July 2026. To the author's knowledge, this represents the first documented observation of the species from Theotokos or the wider southeastern Pelion survey area.

Both individuals occurred within shallow rocky coastal habitat and were associated with refuges provided by the hard substrate.

Follow-up count

The survey conducted on 22 July 2026 recorded a total of 32 visible individuals.

Individuals were not restricted to a single apparent size cohort: small juveniles, intermediate-sized crabs and large specimens were all present. Crabs

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occupied both exposed rock surfaces and sheltered positions within crevices and interstitial spaces beneath and between stones.

A particularly concentrated occurrence was recorded at and around 39.19358894° N, 23.34314461° E, where numerous individuals occupied (Table 1) adjacent refuges within the same section of rocky habitat. This area is interpreted as a probable local aggregation, settlement and refuge site.

No formal measurements of carapace width or length were obtained. No individuals were collected, and sex and reproductive condition were not determined; in particular, ovigerous females were not confirmed during the survey.

Table 1. Summary of observations of *Percnon gibbesi* at Theotokos.

Date	Survey Type	Individuals recorded	Main observation
21 July 2026	Initial biodiversity survey	2	First photographic documentation at Theotokos.
22 July 2026	Targeted follow-up survey	32	Multiple size classes and dense aggregation within rocky refuges.

Discussion

The detection of 32 *P. gibbesi* individuals only one day after the initial observation underscores the importance of prompt, targeted surveys following the discovery of a non-indigenous species. Had monitoring ended after the first two observations, the record might reasonably have been interpreted as an isolated or very recent arrival. The follow-up survey instead revealed a substantially larger local aggregation.

The presence of individuals across several visually distinguishable size classes is consistent with an established or actively establishing population. Earlier studies in Greek waters similarly treated the co-occurrence of multiple size classes as evidence of establishment (Thessalou-Legaki et al., 2006). Size variation alone, however, cannot demonstrate local reproduction: small individuals may equally have arrived through larval recruitment from local adults or from populations elsewhere in the Aegean Sea.

The habitat at Theotokos is consistent with the known Mediterranean ecology of the species. *Percnon gibbesi* commonly occupies shallow rocky bottoms, particularly areas with boulders and stones offering suitable crevices and limited algal turf (Katsanevakis et al., 2010; Thessalou-Legaki et al., 2006; Deudero et al., 2005; Zenone et al., 2016). Studies in the western Mediterranean have reported maximum abundance at approximately 1 m depth and a preference for pebble and boulder substrates (Deudero et al., 2005), while experimental evidence indicates that megalopae rapidly select hard, stable substrates offering both shelter and access to food (Zenone et al., 2016).

The occurrence of juveniles within the rocky interstices at Theotokos may therefore reflect successful settlement in suitable habitat. Confirmation of this interpretation, however, requires repeated observation, formal measurement and, where legally and ethically appropriate, assessment of reproductive condition.

The marked increase from two to 32 recorded individuals within a single day does not necessarily indicate rapid population growth over that interval. The more parsimonious interpretation is that a larger, previously undetected population was already present, and that the initial survey captured only a fraction of the individuals actually occupying the site. The species' flattened morphology, rapid escape response, and tendency to retreat into narrow crevices are known to reduce detectability and can cause visual surveys to substantially underestimate true abundance.

The pathway by which *P. gibbesi* reached southeastern Pelion cannot be determined from the present observations. Regional larval dispersal from established Aegean populations is a plausible mechanism. Vessel-associated transport represents an additional possible pathway, but should not be assumed as the principal mechanism in the absence of information on nearby harbours, anchoring sites, and recreational or commercial vessel activity.

The ecological consequences of this newly documented population remain unknown. *Percnon gibbesi* feeds predominantly on benthic algae, and its flexible feeding ecology likely contributes to its invasion success. Potential ecological interactions may include competition for space or shelter with native crabs, dietary overlap with native grazers, and modification of shallow algal assemblages. These impacts should not be inferred from the present occurrence record alone and warrant dedicated, site-specific ecological investigation.

Monitoring and management recommendations

The Theotokos population warrants reassessment through standardised, repeatable surveys. Recommended actions include the following:

- Establishing fixed belt transects or timed-search stations centred on the initial coordinates.

- Recording survey length, width, depth range, water temperature, substrate type, algal cover and number of available refuges.

• Photographing individuals alongside a scale, or using calibrated underwater imagery, to estimate carapace width.

• Recording abundance separately for juveniles, intermediate-sized individuals and adults.

• Searching specifically for ovigerous females, recently settled juveniles and moults.

• Mapping the alongshore distance over which individuals occur.

• Repeating surveys monthly during the first year to assess persistence, recruitment and seasonal variation in detectability.

• Surveying neighbouring rocky coastal sites to determine whether the occurrence is restricted to Theotokos or reflects a wider southeastern Pelion population.

• Recording native crabs, sea urchins, gastropod grazers, macroalgae and other potentially interacting taxa along the same transects.

Any proposal for physical removal should be grounded in a clearly defined management objective, supported by evidence of feasibility, applicable permits, and a realistic assessment of eradication prospects. Removal of visible specimens without adequate follow-up is unlikely to eliminate the population where hidden individuals, juveniles, or ongoing larval recruitment remain present.

Conclusion

The observations reported here constitute the first documented occurrence of *Percnon gibbesi* at Theotokos and a potentially significant range record for southeastern Pelion. Two individuals were initially documented on 21 July 2026, and a follow-up survey conducted the next day recorded 32 individuals at and around 39.19358894° N, 23.34314461° E.

The co-occurrence of juveniles, intermediate-sized specimens and large individuals strongly suggests an established or actively establishing population rather than the arrival of two isolated, transient crabs. The concentrated use of rocky shelters indicates a probable settlement and refuge site, although current evidence is insufficient to characterise the location as a nest or to confirm local reproduction.

This finding underscores the value of rapid-response field surveys and sustained local monitoring programmes in detecting previously undocumented populations of marine non-indigenous species.

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The author used Claude to rewrite some of the content for grammar issues.

Author contributions

Christos Taklis: conceptualisation, field investigation, species identification, photographic documentation, data collection, manuscript preparation and project administration.

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Conflict of interest

The author declares no conflict of interest.

Ethics and permits

The study was based on non-invasive underwater visual observations and photographic documentation. No animals were captured, handled, injured or removed.

Data availability

Photographs, observation records and associated field information are maintained by Merman Conservation Expeditions Ltd and may be made available upon reasonable request. Verified records may also be submitted to relevant national and European non-indigenous species databases.

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